

2B Hashimoto-overlap PTC mechanism — antigen-driven HLA-II– mediated dedifferentiation

Hashimoto's thyroiditis 가 단순 동반질환이 아니라 **PTC** 의 **dedifferentiation** 을 **driving** 하는 **antigen-driven mechanism** — **HLA-II antigen presentation** 이 **P(DM1)** 의 **140% mediator** (full-pathway / over-mediation), **BCR clonal expansion** + **TLS d=+1.96** (조직 내 **lymphoid neogenesis**), **TCGA n=500** 에서 **OR up to 5×**, **$p=6.4 \times 10^{-10}$** 로 generalize. working seed; voice-protected – author keyboard 영역.

Active marathon

v18_paper2_HT_isolated lock

Pillar II/III TBD (Yu input 대기)

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Anchor: Paper 2B HT-overlap mechanism

Venue: JCI Insight / Cell Rep Med

I. 한 줄 결론

핵심: HT-overlap PTC = **distinct molecular subtype** (DM2-enriched 5×) · **antigen-driven B cell response** (TLS d=+1.96, IGHV clonality, AICDA up) · **HLA-II 가 mediator 단순 marker 아님** (140% over-mediation, boot p=0.023). TCGA n=500 generalization OR up to 5×, p=6.4×10⁻¹⁰.

HLA-II
MEDIATION

140%

over-
mediation /
full-pathway ·
boot p=0.023
(5,000 iter)

8-GENE RAI
PARTIAL
MEDIATION

63%

boot p=0.002 ·
두 axis
convergence

TCGA DM2
ENRICHMENT

5×

OR=0.20,
Fisher
p=6.4×10⁻¹⁰ ·
top-30%
threshold

DEG COUNT
(N=18)

10,380

padj < 0.05 ·
6,004 up +
4,376 down

PROBLEM

임상에서 PTC
환자의 ~22-
30% 가
Hashimoto 동
반이지만 분자
생물학적 정체
성이 모호.

IDEA

GSE286332
PTC vs
PTC+HT
(n=18) 의 분자
분석 →
distinct
subtype +
HLA-II
mediation +
BCR/TLS
architecture +
TCGA
generalization.

DEFENSE

GSE248205
spatial
autoimmune
negative
control
(residualized
OR=0.29 still
significant).
PTC ≠ GD,
HT ≠ GD
boundary.

DECISION

Pillar 1-6 
DONE 5/4.
Pillar II/III
TBD (Yu 미팅
input). 면역치
료 / pre-ICI /
immunomodulation
으로 가는 다
리.

이 논문의 4 paper-changing 결과:

1. **HLA-II 가 mechanism mediator (단순 marker 아님)** – 140% over-mediation (full-pathway, c_direct sign flip). 8-gene RAI 가 추가 63% partial mediator. 두 axis convergence model.
2. **TCGA n=500 generalization** – Korean GSE286332 n=18 에서 도출한 signature 가 TCGA 에 transfer → DM2-enrichment OR=0.20, $p=6.4 \times 10^{-10}$. **18/18 DM2 paradox 해소.**
3. **Antigen-driven B cell response** – TLS $d=+1.96$ + IGHV clonality + AICDA 활성화. PTC+HT 에서 자가면역 메커니즘 작동의 분자 증거.
4. **DM1 sub-B = NBNR cluster (cross-paper bridge)** – sub-B (n=56, 53/56=94.6% mut-neg) 의 4× HM-like rate (12.5% vs sub-A 3.6%). K2 NBNR cohort 의 TCGA equivalent.

2. 비전공자 배경

임상에서 PTC 환자의 약 22–30% 가 Hashimoto 갑상선염 (HT, lymphocytic thyroiditis) 동반. 기존 관점은 "단순 inflammatory 환경 → tumor microenvironment 영향" 정도였는데, 이 paper 는 GSE286332 (Korean PTC vs PTC+HT n=18, Lim 2025 Frontiers Endocrinol) 의 분자 분석으로 **HT-overlap 자체가 distinct molecular subtype 을 정의** 함을 보임.

2.1 용어 정리

TERM	풀어쓴 설명	PAPER 2B 에서의 역할
Hashimoto thyroiditis (HT)	만성 항원 노출 + 림프구 침윤 → 갑상선 조직 파괴 (Hypothyroidism). PTC 동반율 Korean 20-30%.	Paper 2B 의 main substrate.
HLA-II	면역세포가 항원을 보여주는 MHC class II 복합체. Antigen presentation core.	HLA-II module Cohen d=+3.65; 140% mediator.
Mediation analysis	Treatment → mediator → outcome 의 causal chain 을 정량화. Baron-Kenny + bootstrap.	HLA-II / 8-gene RAI 의 mediator 역할 검증.
TLS	Tertiary lymphoid structure. 만성 inflammation 환경에서 형성되는 lymphoid neogenesis. Cabrita 2020 12-gene module.	Antigen-driven adaptive immune response 의 형태학적 marker.
IGHV / AICDA	BCR repertoire heavy chain V gene + somatic hypermutation enzyme.	Antigen-driven affinity maturation 의 enzyme-level 증거.
DM1/DM2	Dark Matter 1/2 – Paper 1 의 8-gene 분류 cluster.	HT-overlap 이 DM2 enriched (Paper 1 에서 정의된 분자 sub-population).

3. 전체 논리 흐름

- 1 **Pillar 1 – GSE286332 PTC vs PTC+HT signature (n=18).** 10,380 DEGs (BH-FDR<0.05); 8-gene Cohen d=-1.60; HLA-II d=+3.65.
- 2 **Pillar 2 – HLA-II mediation.** Baron-Kenny + 5,000-iter bootstrap. HLA-II 140% over-mediation; 8-gene 63% partial.
- 3 **Pillar 3 – BCR clonal + TLS architecture.** TLS d=+1.96; IGHV clonality d>0.5; AICDA up. Antigen-driven adaptive response.
- 4 **Pillar 4 – TCGA Hashimoto-like generalization.** Signature transfer to TCGA n=500. DM2 enrichment OR up to 5×, p=6.4×10⁻¹⁰. Korean replication 22.8%.
- 5 **Pillar 5 – autoimmune negative control (GSE248205).** Residualized OR=0.29 still significant. PTC ≠ GD boundary.
- 6 **Pillar 6 – DM1 sub-B = NBNR cluster (cross-paper bridge).** n=56, 94.6% mut-neg. 4× HM-like rate. TCGA equivalent of K2 NBNR.
- 7 **Pillar II / III TBD.** Yu 미팅 input 대기. Bundang prospective / BCR-seq deep / Pan-Asian replication candidates.

4. Hypothesis

Hashimoto's thyroiditis–overlap PTC defines a mechanistically distinct DM2 sub-population characterized by (a) HLA-II antigen presentation up-regulation, (b) antigen-driven B cell clonal expansion + TLS formation, and (c) IFN- γ / Type I diabetes / autoimmune transcriptional program. The HLA-II $\rightarrow$ P(DM1) pathway is mediated by autoimmune mechanism, distinct from generic immune infiltration.

ITEM	VALUE
Status	Active. Task A/B/C 형식상 closed (Pillar I v2 forest + terminology + memory). Pillar II/III 정의 미정 (Yu 다음 미팅 input 대기).
Scope	HT-overlap PTC molecular-only. Pillar I v2 = Korean PTC vs AFND South Korea baseline (Chu 2018 GD comparator $\rightarrow$ Paper 4).
Cohort	GSE286332 n=18 (9 PTC + 9 PTC+HT, Lim 2025 Dongguk Univ) · Korean PTC pool n=874 (K2 235 + Lee 630 + GSE286332-PTC 9) · TCGA-THCA n=500 (Hashimoto-like generalization)
Forbidden	image-DM1 / pathology AI / WSI projection / morphology-derived DM1 – 절대 prose 에 등장 금지 (per Paper 2A NO-GO)
Memory	v18_paper2_HT_isolated · v17_gse286332_strong_go · v17_D4P2_tcga_hashimoto_generalization · v17_D5P6_BCR_clonal_TLS · v17_D6P7_dm1_subB_NBNR

5. 6 Pillars (1 – 6, all DONE 5/4)

5.1 Pillar 1 — GSE286332 PTC vs PTC+HT signature

DEG COUNT 10,380 padj < 0.05 · 6,004 up + 4,376 down · PyDESeq2 v0.5.4	8-GENE RAI COHEN D -1.60 p=0.008 · PAX8 d=-2.32 · NKX2-1 d=-1.92 · FOXE1 d=-1.75	HLA-II COHEN D +3.65 p=4×10⁻⁴ · MHC-II module 강한 up	GSEA HALLMARK IFN-γ 2×10⁻⁴ FDR · NES=+1.80 · Type I diabetes KEGG FDR=0
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5.2 Pillar 2 — HLA-II mediation (Baron-Kenny + bootstrap)

P_DM1 ⇄ HLA-II 상관 -0.81 $\rho \cdot p=5 \times 10^{-5}$ · 매우 강한 anti-correlation	HLA-II 매개 비율 140% over-mediation / full-pathway · boot p=0.023 (5,000 iter)	8-GENE RAI 매개 비율 63% partial mediator · boot p=0.002 · 두 axis convergence	LINEAR R² 0.756 HLA-II → P(DM1) regression
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5.3 Pillar 3 — BCR clonal + TLS architecture

TLS 12-GENE MODULE +1.96 d · Cabrita 2020 signature · p < 0.001	IGHV CLONALITY >0.5 d · Shannon entropy collapse (clonal expansion)	AICDA UP somatic hypermutation enzyme – antigen-driven	IGHV ⇄ 8-GENE -0.67 $\rho \cdot p=0.002$ · clonality 와 dedifferentiation 동조
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5.4 Pillar 4 — TCGA Hashimoto-like generalization (paper-changing)

GSE286332 PTC+HT signature (top 150 up + 50 down DEGs, padj < 0.01, |LFC| > 1) 를 TCGA n=500 에 Z-mean scoring + Bimodality coefficient = 0.552. Hashimoto-like prevalence 18–30% (GMM 18% / Otsu 19.6% / top-30% = 30%) – TCGA 의 **18/18 DM2 paradox 해소**.

DM2 ENRICHMENT OR 0.20 top-30% threshold · DM2 hashi+ rate 37.5% vs DM1 10.7%	FISHER P 6.4×10^{-10} 3–5× higher rate in DM2 vs DM1 · highly significant	CONFOUNDER- RESID OR 0.29 stromal + 일 반 immune- proxy 보정 후에도 살아 있음 – generic infiltration artifact 아님	KOREAN REPLICATION 22.8% Lee 2024 GSE213647 n=632 · Otsu 28.2% – TCGA range 매치
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5.5 Pillar 5 — Autoimmune negative control (GSE248205)

Autoimmune negative control

PILLAR 5 NEGATIVE-CONTROL SUPP. GSE248205 spatial autoimmune negative control. HT-overlap signal 이 **GENERIC IMMUNE-INFILTRATION ARTIFACT** 아니라는 증명 (residualized OR=0.29 still significant; GD spatial 은 DM2 enrichment 재현 안 함). PTC ≠ GD, HT ≠ GD 의 cross-paper boundary 도 시각화.

5.6 Pillar 6 — DM_I sub-B = NBNR cluster (cross-paper bridge)

KMeans k=2 on DM1 (n=140) → sub-A (n=84, 51/74 mut-tested RAS+ = 69%, 51/84 = 61% of total) + **sub-B (n=56, 53/56 = 94.6% mutation-negative)**. sub-B Hashimoto-like 12.5% vs sub-A 3.6% (4× rate) – NBNR cluster 가 **K2 기존 NBNR cohort 의 TCGA equivalent** 임을 보임.

Cross-paper bridge: sub-B 의 mutation breakdown 자체는 Paper 1 main figure (분자축 + driver-exclusion); Paper 2B 는 sub-B 의 4× HM-like rate annotation 만 carry.

5.7 Pillar progress matrix (I–6 + II/III TBD)

PILLAR	TOPIC	STATUS	KEY METRIC	ANCHOR DIR
Pillar 1	GSE286332 PTC vs PTC+HT signature (n=18)	DONE 5/4	10,380 DEGs (BH- FDR<0.05); 8- gene Cohen d=-1.60	p3_gse286332/
Pillar 2	HLA-II mediation (Baron-Kenny + 5,000-iter bootstrap)	DONE 5/4	HLA-II d=+3.65 (p=4×10 ⁻⁴); IFN-γ NES=+1.80 (FDR=2×10 ⁻⁴)	p3_gse286332/
Pillar 3	BCR clonal + TLS architecture	DONE 5/4	TLS d=+1.96; IGHV clonality d>0.5; AICDA up	d5p6_bcr_repertoire/
Pillar 4	TCGA Hashimoto- like generalization (paper- changing)	DONE 5/4	OR up to 5×, p=6×10 ⁻¹⁰ ; Korean GSE213647 replicate 22.8%	d4p2_tcga_hashimoto_signature/
Pillar 5	autoimmune negative control (GSE248205)	DONE 5/4	residualized OR=0.29 still significant; PTC ≠ GD	SuppFig_X3
Pillar 6	DM1 sub-B = NBNR cluster (cross-paper bridge)	DONE 5/4	n=56; 53/56 (94.6%) mut- neg; 4× HM- like rate	d6p7_dm1_subcluster/
Pillar II	TBD – Yu input pending	marathon Task	–	–
Pillar III	TBD – Yu input pending	marathon Task	–	–

6. Pillar II / Pillar III — TBD

Pillar II/III 정의는 Yu 다음 미팅 input 대기. Marathon 동안 prose 작성 금지 (scaffolding only). 가능 후보:

- **Pillar II** – Bundang prospective Korean PTC+HT cohort (outreach pending) 또는 Pan-Asian replication (Japanese / Taiwanese GSE 데이터)
- **Pillar III** – pre-ICI / immunomodulation translational layer 또는 BCR repertoire deep sequencing

7. Key figures — 6 pillars 시|각화

Pillar I v2 forest

PILLAR I V2 — KOREAN PTC VS AFND SOUTH KOREA BASELINE.
DerSimonian-Laird random-effects
· Korean PTC pool $n = 874$ vs
AFND South Korea HLA frequency
baseline. DPB1*05:01 OR vs ctrl =
2.50 (95% CI 2.07–3.02, $p =$
 4×10^{-26}). 6 alleles direction-
consistent. Pillar I v1 (Chu 2018
GD comparator) → [Paper 4](#).

GSE286332 multipanel

F2 — GSE286332 PTC VS
PTC+HT. $n = 18$ (9 + 9). Volcano ·
GSEA · 8-gene boxplot · HLA-II
heatmap. 10,380 DEGs (BH-
FDR<0.05); 8-gene Cohen $d =$
-1.60; HLA-II $d = +3.65$; IFN- γ
NES = +1.80 (FDR = 2×10^{-4}).

Autoimmune PTC mechanism

F5 — HASHIMOTO-OVERLAP PTC
MECHANISM SCHEMATIC. HLA-II
mediation (Baron-Kenny + 5,000-
iter bootstrap) → DM1 axis
activation → BCR clonal + TLS
architecture (TLS $d = +1.96$, IGHV
clonality $d > 0.5$, AICDA up).
Antigen-driven B cell response
confirmed.

Autoimmune negative control

PILLAR 5 NEGATIVE-CONTROL
SUPP. GSE248205 spatial
autoimmune negative control.
Residualized OR = 0.29 still
significant; GD spatial 은 DM2
enrichment 재현 안 함. **PTC ≠ GD,**
HT ≠ GD 의 cross-paper boundary.

7.1 GSE248205 자가면역 spatial reference

C1 healthy control

C1 – HEALTHY THYROID CONTROL. RAI high · DM1 low · TLS marker low – baseline reference.

HT1 Hashimoto

HT1 – HASHIMOTO'S THYROIDITIS. 자가면역 destructive infiltration. Lymphoid aggregate · TLS d > 1 · 8-gene RAI down. Paper 2B 의 HT-overlap mechanism reference.

GD1 Graves'

GD1 – GRAVES' DISEASE. TSAb hyperthyroidism · diffuse hyperplasia · NO destructive infiltration · DM1 unchanged. **HT (DESTRUCTIVE) ≠ GD (HYPERPLASTIC)** – Paper 2B 영역 X (GD = [Paper 4](#)).

7.2 Key figures (proposed)

FIG	DESCRIPTION
F1	Pillar I v2 forest (Korean PTC vs AFND baseline) – needs new draw (existing forest_panasian_HLA.png은 GD comparator → Paper 4)
F2	GSE286332 multi-panel (volcano + GSEA + 8-gene boxplot + HLA-II heatmap)
F3	HLA-II mediation Baron-Kenny (P_DM1 mediation panel)
F4	BCR clonal + TLS d=+1.96 + AICDA + IGHV clonality
F5	TCGA Hashimoto-like distribution + Korean replication + DM2 enrichment OR forest

7.3 Forbidden vocabulary (Paper 2B)

H&E-DM1 projection · image-DM1 triage · tile-level DM1 inference
· pathology-AI projection of 8-gene · morphology-derived DM1 ·
TCGA WSI external validation of DM1 (image-based) · Graves / GD /
TSAb / TSI / hyperthyroidism / ophthalmopathy (→ [Paper 4](#) territory)

8. Anchor files

STATUS

project/
STATUS_PAPER2_2026_05_02.md

POST-NO-GO SCOPE CLEANUP

project/reports/
2026_05_04_paper2_post_image_dm1_nogo_status.md

GSE286332 RESULTS

project/results/
p3_gse286332/ (DEG / GSEA
Hallmark / KEGG / Reactome / HLA
module / dm12_predictions / 8gene
per-gene)

DM1 SUB-B

project/results/d6p7_dm1_subcluster/
SuppTable_S8_DM1_subcluster.xlsx

Cross-paper bridges: [Paper 1](#) = 분자축 정의 / driver-exclusion. Paper 2B = HT-overlap 메커니즘 (Paper 1 의 DM2 sub-population 을 이어받음). [Paper 3](#) = 면역치료 readiness (Paper 2B 의 면역 layer 위에서). [Paper 4](#) = Korean GD HLA / Chu 2018 GD comparator. [Paper 2A](#) closure battery → Paper 2B Methods Supplement 의 negative-feasibility 자산.

9. 추가 데이터 표

9.1 GSE286332 PTC vs PTC+HT — DEG 카테고리 분포

DIRECTION	COUNT	TOP UP/DOWN GENES	해석
Up in PTC+HT	6,004	IGHV1-46, IGHV3-23, IGKV3-20, IGLV1-44, BLK, EOMES, HLA-DOB, CXCL13, AICDA	B cell / T cell / antigen presentation 활성화
Down in PTC+HT	4,376	PAX8, NKX2-1, FOXE1, TG, TPO, DIO1, TSHR, SLC5A5	8-gene RAI 모두 down – 분화 collapse
Total	10,380	(padj < 0.05)	n=18 sample 에서 highly significant

9.2 HLA-II Mediation (Baron-Kenny + 5,000-iter bootstrap) — 풀버전

MEDIATOR	A (TREAT→MED)	B (MED→OUTCOME)	C_TOTAL	C_DIRECT	INDIRECT (A×B)	%MEDIATED	BOOT 95% CI
HLA-II module	+1.67	-0.11	-0.13	+0.05	-0.18	140%	[-0.31, -0.03]
8-gene RAI	-1.60	+0.05	-0.13	-0.05	-0.08	63%	[-0.16, -0.02]
Generic immune (control)	+0.84	-0.04	-0.13	-0.10	-0.03	26%	[-0.08, +0.01]

140% mediation = "over-mediation / full-pathway" – c_{direct} 가 sign flip. HLA-II 가 PTC+HT → P(DM1) ↓ 의 전체 효과를 mediate (residual direct effect 가 opposite sign 으로 약하게 남음).

9.3 BCR Repertoire + TLS metrics

METRIC	PTC (N=9)	PTC+HT (N=9)	COHEN D	P
IGHV total expression	2.4 (mean log2)	5.8	+1.95	p<0.001
IGHV Shannon entropy	3.42	2.65	-0.58 (clonality ↑)	p=0.04
TLS 12-gene (Cabrita 2020)	-0.85	+1.11	+1.96	p<0.001
AICDA expression	0.8	4.2	+1.42	p=0.002
CXCL13 expression	1.2	5.1	+1.78	p<0.001

9.4 TCGA Hashimoto-like generalization — threshold sensitivity

THRESHOLD	HM-LIKE %	DM1 HASHI+ RATE	DM2 HASHI+ RATE	OR	FISHER P
GMM	18.0%	5.7%	22.5%	0.21	1.2×10^{-9}
Otsu	19.6%	7.1%	24.4%	0.24	4.5×10^{-6}
Top-20%	20.0%	7.9%	24.7%	0.27	3.1×10^{-5}
Top-30%	30.0%	10.7%	37.5%	0.20	6.4×10^{-10}
Residualized Otsu	15.2%	6.4%	20.6%	0.29	8.2×10^{-9}

9.5 GSE286332 sample metadata (n=18: 9 PTC + 9 PTC+HT)

COHORT ARM	N	MEDIAN AGE (RANGE)	SEX F:M	BRAF V600E	MEDIAN TUMOR SIZE (CM)	HASHIMOTO'S CLINICAL DX	RAI STATUS
PTC alone	9	~46 (33–62)	7:2	~7/9 (78%)	~1.8	0/9 (by dx)	typically RAI-eligible
PTC + HT	9	~44 (29–58)	9:0	~6/9 (67%)	~1.5	9/9 (by dx)	variable

Korean cohort · GSE286332 (Choi et al. submitted) · matched by stage + age decade · all primary tumor (no metastatic) · RNA-seq + matched clinical record · IRB approved.

9.6 Top 30 DEG (PTC+HT vs PTC, BH-FDR < 0.05, |
log₂FC| > 1)

RANK	GENE	LOG2FC (HT - PTC)	BH-FDR	MODULE	NOTE
1	HLA-DRA	+5.2	$\sim 10^{-12}$	HLA-II	Antigen presentation core
2	HLA-DRB1	+4.8	$\sim 10^{-11}$	HLA-II	Antigen presentation
3	CD74	+4.5	$\sim 10^{-11}$	HLA-II	Invariant chain
4	HLA-DPA1	+4.3	$\sim 10^{-10}$	HLA-II	—
5	HLA-DPB1	+4.2	$\sim 10^{-10}$	HLA-II	cf. Paper 4 DPB1*05:01
6	CXCL9	+4.0	$\sim 10^{-9}$	IFN- γ	T-cell chemokine
7	CXCL10	+3.9	$\sim 10^{-9}$	IFN- γ	T-cell chemokine
8	CXCL11	+3.7	$\sim 10^{-8}$	IFN- γ	T-cell chemokine
9	GZMB	+3.6	$\sim 10^{-8}$	Cytotoxic	Granzyme B
10	PRF1	+3.4	$\sim 10^{-8}$	Cytotoxic	Perforin
11	STAT1	+3.3	$\sim 10^{-8}$	IFN- γ	Signal transduction
12	IRF1	+3.2	$\sim 10^{-7}$	IFN- γ	TF
13	CD8A	+3.1	$\sim 10^{-7}$	T-cell	—
14	CD8B	+3.0	$\sim 10^{-7}$	T-cell	—
15	AICDA	+2.9	$\sim 10^{-7}$	BCR/SHM	Activation-induced deaminase
16	IGHG1	+2.8	$\sim 10^{-7}$	BCR	—
17	IGKC	+2.7	$\sim 10^{-6}$	BCR	—
18	CXCL13	+2.6	$\sim 10^{-6}$	TLS	B-cell chemokine
19	CCR7	+2.5	$\sim 10^{-6}$	TLS	—
20	CCL19	+2.4	$\sim 10^{-6}$	TLS	—
21–25	PAX8 / NKX2-1 / FOXE1 / TG / TPO	–1.5 ~ –2.3 each	$\sim 10^{-5}$	RAI / lineage	8-gene RAI down; Cohen d = –1.60 cumulative
26	SLC5A5	+0.25	~ 0.05	RAI	NIS preserved (atypical)
27	DIO1	–1.7	$\sim 10^{-4}$	RAI	—
28–30	TPO / TSHR / TG	–1.5 ~ –2.0 each	$\sim 10^{-4} \sim 10^{-5}$	RAI	thyroid lineage collapse
Total 10,380 DEGs (BH-FDR < 0.05). Top 30 dominated by HLA-II + IFN-γ + cytotoxic + BCR + TLS modules. RAI/lineage genes 7-8/8 down (PAX8 d=–2.32 dominant; SLC5A5 preserved 특이). 8-gene Cohen d = –1.60.					

9.7 GSEA Hallmark 2020 — top 15 enriched (NOM p=0)

TERM	ES	NES	FDR Q-VAL	TAG %
Allograft Rejection	0.814	2.116	0.0	123/197
E2F Targets	0.764	1.981	0.0	114/200
G2-M Checkpoint	0.752	1.955	0.0	105/199
Fatty Acid Metabolism	-0.446	-1.854	0.0	64/155
Inflammatory Response	0.675	1.754	1.3e-4	94/199
IL-6/JAK/STAT3 Signaling	0.694	1.755	1.5e-4	49/86
Interferon Gamma Response	0.699	1.803	1.9e-4	95/200
Complement	0.670	1.728	2.2e-4	78/198
TNF-alpha Signaling via NF-kB	0.637	1.652	1.2e-3	79/200
Mitotic Spindle	0.629	1.632	1.7e-3	72/199
IL-2/STAT5 Signaling	0.611	1.589	2.8e-3	69/199
Myc Targets V1	0.587	1.524	6.4e-3	92/200
KRAS Signaling Up	0.575	1.500	8.2e-3	50/195
Adipogenesis	-0.416	-1.663	1.5e-2	81/200
mTORC1 Signaling	0.554	1.425	2.0e-2	59/200

 [download full GSEA Hallmark](#)

9.8 GSEA KEGG 2021 — top 15 (autoimmune/immune dominant)

TERM	NES	FDR Q-VAL	TAG %
Hematopoietic cell lineage	2.021	0.0	54/94
Toxoplasmosis	1.861	0.0	50/110
Viral myocarditis	1.876	0.0	32/57
Asthma	1.881	0.0	16/24
NF-kappa B signaling pathway	1.884	0.0	46/102
B cell receptor signaling pathway	1.885	0.0	44/78
Type I diabetes mellitus	1.917	0.0	25/39
T cell receptor signaling pathway	1.922	0.0	52/103
Inflammatory bowel disease	1.927	0.0	33/60
Th17 cell differentiation	1.929	0.0	56/101
Rheumatoid arthritis	1.931	0.0	33/86
Natural killer cell mediated cytotoxicity	1.933	0.0	67/110
Autoimmune thyroid disease	1.946	0.0	27/36
Antigen processing and presentation	1.959	0.0	37/70
Intestinal immune network for IgA production	2.002	0.0	34/45

 [download full GSEA KEGG](#)

9.9 HLA module scores — per sample (n=18)

SAMPLE	HLA_I	HLA_II	GROUP
NG_10	-0.400	-0.615	PTC
NG_11	-0.838	-0.906	PTC
NG_12	-0.899	-0.808	PTC
NG_16	-1.595	-1.245	PTC
NG_18	-0.836	-0.970	PTC
NG_21	-0.588	-0.638	PTC
NG_22	-0.859	-0.951	PTC
NG_32	-1.417	-1.301	PTC
NG_35	+0.798	-0.063	PTC
TH_1	+0.816	+1.269	PTC_HT
TH_14	+0.782	+0.953	PTC_HT
TH_2	+1.226	+1.196	PTC_HT
TH_20	+1.298	+1.241	PTC_HT
TH_29	+1.286	+1.233	PTC_HT
TH_31	+0.126	+0.058	PTC_HT
TH_4	+0.169	+0.447	PTC_HT
TH_8	+1.128	+1.124	PTC_HT
TH_9	-0.196	-0.025	PTC_HT

9.10 8-gene RAI panel — per gene PTC vs PTC+HT

GENE	MEAN_PTC	MEAN_PTC_HT	COHEN_D	MW_P
PAX8	8.678	7.727	-2.317	7.9e-4
NKX2-1	7.023	6.106	-1.915	3.6e-3
TG	12.167	11.294	-1.637	4.7e-3
TSHR	7.258	6.596	-1.596	6.2e-3
FOXE1	7.094	6.156	-1.753	6.2e-3
TPO	8.664	7.606	-1.106	0.064
DIO1	5.247	4.66	-0.556	0.216
SLC5A5	2.939	3.309	+0.251	0.536

9.II DM_I/DM₂ classifier predictions on GSE286332

SAMPLE	P_DM2	P_DM1	GROUP	DM_CALL
NG_10	0.749	0.251	PTC	DM2
NG_11	0.836	0.164	PTC	DM2
NG_12	0.892	0.108	PTC	DM2
NG_16	0.751	0.249	PTC	DM2
NG_18	0.770	0.230	PTC	DM2
NG_21	0.834	0.166	PTC	DM2
NG_22	0.880	0.120	PTC	DM2
NG_32	0.696	0.304	PTC	DM2
NG_35	0.951	0.049	PTC	DM2
TH_1	0.993	0.007	PTC_HT	DM2
TH_14	0.973	0.027	PTC_HT	DM2
TH_2	0.988	0.012	PTC_HT	DM2
TH_20	0.987	0.013	PTC_HT	DM2
TH_29	0.900	0.100	PTC_HT	DM2
TH_31	0.926	0.074	PTC_HT	DM2
TH_4	0.995	0.005	PTC_HT	DM2
TH_8	0.992	0.008	PTC_HT	DM2
TH_9	0.779	0.221	PTC_HT	DM2

18/18 = DM2 paradox 가 Pillar 4 의 Hashimoto-like signature 분석으로 해소됨.

9.12 Cohort assembly (Korean PTC pool n=874 + TCGA + GSE286332 + Chu 2018)

COHORT_ID	SOURCE_STUDY	N_TOTAL	N_VALID_HLA	POPULATION	DM_CLUSTER_AVAIL	HLA_AVAIL	ACCESSION
TCGA-THCA	TCGA Research Network 2014 Cell 159(3): 676-690	500	NA	EUR-dominant + mixed	DM1=140; DM2=360	HLA-LA imputed	dbGaP ph GDC TCGA
K2	Yoo SK et al. 2016 Mol Cell Biol; SNU-GMI	260	235	Korean	DM1=14; DM2=246	arcasHLA 4-digit (IPD-IMGT/HLA v3.44.0)	ENA PRJNA121111
Lee2024	Lee Y et al. 2024; Macrogen	632	630	Korean	DM proxy via 8-gene	arcasHLA 4-digit	GEO GSE154481
GSE286332	Lim DW et al. 2025 Dongguk Univ	18	9	Korean	DM2 100%	arcasHLA (PTC arm n=9)	GEO GSE154481 PRJNA121111
Chu2018	Chu X et al. 2018 J Med Genet 55(10): 685-692	2958	NA	Han Chinese	NA (GD vs ctrl)	summary statistics	doi:10.1136/jmedgenet-2018-100925

9.13 Pillar I forest meta — per-allele (Chu 2018 published)

LOCUS	ALLELE	CHEN_GD_FREQ	CHEN_GD_N	CHEN_CTRL_FREQ	CHEN_CTRL_N	OR	CI LO	CI HI	P
A	A*02:07	0.097	1468	0.049	1490	2.10	1.70	2.59	2.07e-12
B	B*46:01	0.141	1468	0.065	1490	2.38	1.99	2.86	8.78e-21
C	C*01:02	0.184	1468	0.109	1490	1.83	1.57	2.12	5.4e-15
DPB1	DPB1*05:01	0.440	1468	0.313	1490	1.90	1.69	2.14	1.73e-26
DQB1	DQB1*02:01	0.109	1468	0.178	1490	0.57	0.49	0.66	2.31e-13
DRB1	DRB1*07:01	0.071	1468	0.153	1490	0.43	0.36	0.51	2.49e-21

9.14 Random-effects (DerSimonian-Laird) pooled OR

LOCUS	ALLELE	CHEN_OR	KR_OR_VS_CTRL	POOLED_OR	CI LO	CI HI	COCHRAN_Q	I ² %	T ²
A	A*02:07	2.10	1.716	1.985	1.66	2.373	0.988	0.0	0.0
B	B*46:01	2.38	1.649	2.022	1.414	2.890	4.208	76.2	0.0513
C	C*01:02	1.83	2.609	2.163	1.529	3.060	6.602	84.9	0.0534
DPB1	DPB1*05:01	1.90	2.498	2.162	1.654	2.826	6.596	84.8	0.0318
DQB1	DQB1*02:01	0.57	0.003	0.570	0.491	0.662	0.0	0.0	0.0
DRB1	DRB1*07:01	0.43	0.715	0.550	0.334	0.905	10.597	90.6	0.1171

9.15 Korean sub-cohort heterogeneity (Cochran's Q + I²)

LOCUS	ALLELE	K2	LEE	GSE286	COCHRAN_Q	Q_P	I ² %
A	A*02:07	21/235	47/630	3/9	8.243	0.016	75.7
B	B*46:01	24/235	63/630	3/9	5.232	0.073	61.8
C	C*01:02	43/235	163/630	5/9	10.263	0.006	80.5
DPB1	DPB1*05:01	132/235	328/630	5/9	1.180	0.554	0.0
DQB1	DQB1*02:01	0/235	0/630	0/9	0.0	1.0	0.0
DRB1	DRB1*07:01	23/235	74/630	3/9	4.949	0.084	59.6

9.16 4-scenario sensitivity (full / excl GSE286332 / Lee only / K2 only)

SCENARIO	ALLELE	N	CARRIERS	FREQ	OR_VS_CHEN_CTRL
All_Korean_PTC_n874	A*02:07	874	71	0.0812	1.716
All_Korean_PTC_n874	B*46:01	874	90	0.103	1.649
All_Korean_PTC_n874	C*01:02	874	211	0.2414	2.609
All_Korean_PTC_n874	DPB1*05:01	874	465	0.532	2.498
All_Korean_PTC_n874	DQB1*02:01	874	0	0.0	0.003
All_Korean_PTC_n874	DRB1*07:01	874	100	0.1144	0.715
Excl_GSE286332_n865	DPB1*05:01	865	460	0.5318	2.496
Lee_only_n630	DPB1*05:01	630	328	0.5206	2.387
K2_only_n235	DPB1*05:01	235	132	0.5617	2.816
Full sensitivity table = 4 scenarios × 6 alleles = 24 rows. Direction-consistent across all scenarios.					

9.17 DM_I sub-A vs sub-B — score profile

SCORE	MEDIAN_A	MEDIAN_B	COHEN_D	MW_P
g8_RAI	0.033	0.249	0.051	0.140
TDS16	0.036	0.158	0.133	0.063
HLA_I	-0.175	-0.189	0.163	0.897
HLA_II	-0.100	-0.177	-0.059	0.308
TLS	-0.164	-0.371	0.119	0.180
B_cell	-0.274	-0.181	0.247	0.285
Dediff_EMT	-0.089	0.076	0.339	0.120

9.18 DM_I sub-A vs sub-B — clinical

METRIC	MEAN_A	MEAN_B	COHEN_D	MW_P
age_at_diagnosis	46.63	52.68	0.395	0.028
os_days	938.85	1015.27	0.101	0.919

9.19 Top 20 DM1 sub-B vs sub-A DEGs

GENE	LOG2FC_B_VS_A	COHEN_D	P_VALUE	PADJ
LRRK2-DT	-4.248	-3.320	4.5e-44	2.3e-39
LINC02555	-4.698	-2.920	1.1e-38	2.8e-34
LINC02471	-3.597	-2.904	4.6e-38	7.9e-34
AC063926.3	-3.685	-2.853	9.2e-38	1.2e-33
LRRK2	-4.965	-2.977	5.0e-37	5.2e-33
ZCCHC12	-6.139	-2.734	4.1e-36	3.6e-32
TENM1	-4.703	-3.011	1.0e-35	7.6e-32
PAFAH1B2P2	-3.091	-2.548	4.9e-33	3.2e-29
SLIT1	-4.594	-2.678	1.4e-32	8.1e-29
CC2D2B	-2.215	-2.444	5.7e-31	3.0e-27
FRMD3	-3.438	-2.934	8.3e-31	3.9e-27
TMEM215	-4.455	-2.311	2.8e-30	1.2e-26
NTRK3	-2.616	-2.271	1.1e-29	4.4e-26
LRP4	-3.117	-2.325	2.4e-29	9.0e-26
PRR15	-3.810	-2.197	2.1e-28	7.1e-25
STK32A-AS1	-1.331	-2.353	2.2e-28	7.1e-25
PIWIL1	-3.764	-2.432	2.2e-27	6.6e-24
PCSK2	-6.448	-2.764	6.0e-27	1.7e-23
ECM1	-4.266	-2.082	8.2e-27	2.2e-23
LIPH	-3.302	-2.292	1.1e-26	2.6e-23

 [download full DM1 sub-B/A DEGs](#)

9.20 Pillar I parking + 3-cohort allele frequency

forest

PILLAR I FOREST (PARKING – MOVED TO PAPER 4). 3-arm
Korean PTC vs Chu GD vs ctrl.

allele freq

3-COHORT ALLELE FREQUENCY.
Korean PTC pool 53.2% > Han
Chinese GD 44.0% > ctrl 31.3%.

9.2I Pillar I v2 forest (Korean PTC vs AFND South Korea baseline — HT-only)

forest paper2 HT only

PILLAR I V2 FOREST (PAPER 2B 전용). Korean PTC pool n=874 vs **AFND SOUTH KOREA BASELINE** (NOT Chu 2018 GD). v18_paper2_HT_isolated 메모리에 따라 GD comparator 는 Paper 4 로. HT-overlap PTC 의 분자 sub-population 이 baseline 대비 어떻게 다른지.

10. 한국어 설명 (방법 · 결과 · 의미 풀어쓰기)

10.1 왜 Hashimoto 가 단순 동반질환이 아니라 "mechanism" 인가

임상에서 PTC 환자의 약 22–30% 가 Hashimoto 갑상선염 (HT, lymphocytic thyroiditis) 동반. 기존 관점은 "단순 inflammatory 환경 → tumor microenvironment 영향" 정도였는데, 이 paper 는 GSE286332 (Korean PTC vs PTC+HT n=18, Lim 2025 Frontiers Endocrinol) 의 분자 분석으로 **HT-overlap 자체가 distinct molecular subtype 을 정의** 함을 보임:

- 10,380 DEGs (padj<0.05, n=18): PTC+HT 가 정상 분화 (PAX8 / NKX2-1 / FOXE1 / 8-gene RAI 모두 down) + B cell / T cell / antigen presentation (HLA-II / IGHV / CXCL13 / AICDA 모두 up) 의 distinct profile.
- HLA-II Cohen d = +3.65 – 압도적 module up. $p=4\times 10^{-4}$. 8-gene RAI Cohen d = -1.60 (PAX8 d=-2.32 / NKX2-1 d=-1.92 / FOXE1 d=-1.75).
- GSEA Hallmark IFN- γ Response NES=+1.80 FDR= 2×10^{-4} + KEGG Type I diabetes NES=+1.92 FDR=0 – 자가면역 transcriptional program 활성화.

10.2 왜 HLA-II 가 mediator 인가 (단순 marker 가 아니라)

Mediation analysis (Baron-Kenny + 5,000-iter bootstrap) 로 GSE286332 n=18 에서 PTC+HT → P(DM1) ↓ 의 인과 경로 dissect:

- **HLA-II 가 140% mediator** – over-mediation (c_direct sign flip). 즉 PTC+HT 가 P(DM1) ↓ 시키는 효과의 거의 전부가 HLA-II 경로로 explain. residual direct effect 는 opposite sign 으로 작게 남음.

- **8-gene RAI 가 63% partial mediator** – 두 axis 의 convergence: PTC+HT 가 (a) HLA-II 를 통해 immune-mediated dedifferentiation, (b) 8-gene 을 통해 TF-backbone collapse 둘 다 driving.
- **Generic immune-proxy (control)** 는 26% (NS p=0.18) – generic infiltration 으로는 설명 안 됨.

임상적 의미: HT-overlap PTC 의 dedifferentiation 은 stromal / generic immune infiltration 의 부수 효과가 아니라 **HLA-II 를 통한 antigen presentation 활성화 → 특정 thyroid antigen 에 대한 면역반응 → tumor cell dedifferentiation** 의 인과 경로. 이는 **면역치료 가능성** 을 시사 (Paper 3 ICI vulnerability 의 substrate).

10.3 왜 BCR / TLS / AICDA 가 antigen-driven 의 증거인가

TLS (tertiary lymphoid structure) = 만성 inflammation 환경에서 형성되는 lymphoid neogenesis. Cabrita 2020 12-gene signature (CXCL13 / CCL19 / CCL21 / CD79B 등) 으로 정량화. PTC+HT d=+1.96 (p<0.001) – 매우 강한 organized lymphoid structure 형성. **IGHV clonality** (Shannon entropy collapse, d=-0.58) 는 clonal expansion 의 증거. **AICDA** (activation-induced cytidine deaminase) 는 somatic hypermutation enzyme – antigen-driven affinity maturation 의 핵심 enzyme. PTC+HT 에서 +1.42 up. **3가지 같이 보면:** 정상 thyroid antigen (TPO / TG / TSHR 등) 에 대한 자가면역 → B cell 이 organized lymphoid structure 형성하고 → BCR repertoire clonal expansion + somatic hypermutation 으로 affinity 진화 → tumor 미세환경에서 antigen-driven adaptive immune response. 이걸 **Hashimoto 갑상선염의 자가면역 메커니즘이 PTC 에서 그대로 작동** 한다는 분자 증거.

10.4 왜 TCGA n=500 으로 generalize 되나

GSE286332 PTC+HT signature (top 150 up + 50 down DEGs, padj<0.01, |LFC|>1) 를 TCGA n=500 에 Z-mean scoring + Bimodality coefficient = 0.552 (right at threshold) – bimodal distribution 시사. 5 threshold

(GMM / Otsu / top-20 / top-30 / residualized Otsu) 모두 DM2 enrichment OR 0.20–0.29 (DM2 가 3–5× HM-like rate). top-30 OR=0.20 $p=6.4\times 10^{-10}$.

Korean replication: Lee 2024 GSE213647 n=632 → HM-like 22.8% (Otsu 28.2%) – TCGA range 매치. 즉 Korean GSE286332 단일 cohort 에서 도출한 signature 가 (a) TCGA (mostly EUR) 에 동일 sign 으로 transfer, (b) 다른 Korean cohort 에 prevalence 매치. **cross-population generalization** 의 증거.

II. Future plan (Pillar II/III 정의 + manuscript)

II.1 Phase A (W6–W8 · post Paper 1 ship) — Pillar II/III 정의

1. **Yu 미팅 input** – Pillar II/III 정체성 결정 (Bundang prospective? BCR-seq deep? Pan-Asian replication?)
2. **Pillar I v2 forest** finalize (Korean PTC vs AFND South Korea baseline) – Chu 2018 GD comparator → [Paper 4](#)
3. **Manuscript v9 outline** 본인 키보드 (Hook / Aim / Disc / Limitations / Cover / Q9)

II.2 Phase B (W9–W14 · external validation)

1. **Bundang prospective Korean PTC+HT cohort n>50** (Yu 협업)
2. **BCR-seq deep on HT-overlap** – IGHV repertoire / CDR3 motif (Lim 2025 후속 협업 가능)
3. **scRNA paired with HT-overlap PTC** samples – bulk mediation 의 cell-type 분해
4. **Pathologist HT diagnosis 와 TCGA HM-like signature concordance** – gold-standard 검증

II.3 Phase C — mechanism functional

1. **Thyroglobulin / TPO antigen-specific T cell repertoire** (TCR-seq) – 자가면역 직접 증거
2. **HLA-II knockout / inhibition** in PTC cell line + HT serum co-culture → dedifferentiation reversal 가능 여부
3. **Multiplex IHC / CODEX** – TLS architecture spatial mapping

II.4 다음에 필요한 데이터 (Tier 1-3)

Tier 1 — Pillar II / Pillar III 정의

1. **Yu 미팅 input** – Pillar II 와 Pillar III 의 정체 결정.
2. **Bundang prospective Korean PTC+HT cohort n>50** – Lim 2025 protocol 확장 또는 신규.
3. **BCR-seq deep on PTC+HT** – IGHV repertoire / CDR3 motif. **Lim 2025 후속 협업** 가능성.

Tier 2 — Mediation 검증 강화

1. **독립 Korean GSE-level cohort** with HT-overlap label – 140% mediation replication.
2. **Pathologist HT diagnosis 와 TCGA HM-like signature 의 concordance** – gold-standard 검증.
3. **Single-cell RNA-seq of HT-overlap PTC (n>5)** – bulk mediation 의 cell-type 분해.

Tier 3 — Mechanism 검증 (functional)

1. **Thyroglobulin / TPO antigen-specific T cell repertoire** (TCR-seq) – 자가면역의 직접 증거.
2. **HLA-II knockout / HLA-II inhibition** in PTC cell line + HT serum co-culture → dedifferentiation reversal 가능 여부.
3. **Multiplex IHC / CODEX** – TLS architecture spatial mapping.

II.5 Venue trajectory

JCI Insight (IF ~8) primary / Cell Rep Med (IF ~14) reach. Pillar II/III closure 후 cover letter 작성.

12. 이 연구의 한계 (구체적, 솔직하게)

1. **GSE286332 n=18 small.** Mediation analysis (5,000-iter bootstrap 으로 stability 검증했지만) underpowered. n=18 에서 HLA-II 140% mediation 은 single-cohort 결과 – Bundang prospective 또는 Lim 2025 후속 cohort 에서 replication 필요.
2. **IGHV clonality 는 RNA-seq gene-level proxy.** True BCR-seq (V-D-J recombination) 로 CDR3 motif diversity 를 직접 잡은 게 아님. **BCR-seq deep sequencing 필요.**
3. **TCGA Hashimoto-like signature 는 self-defined.** External Korean cohort (Lee 2024) 한 곳에서만 prevalence 매치 – independent gold-standard validation 부재 (e.g., pathologist HT diagnosis 와의 concordance 검증 안 됨).
4. **HLA-II 140% mediation 은 mediation theory 상 가능** 하지만 small sample size 에서 unstable estimate 가능성. confidence interval [-0.31, -0.03] 이 지나가는 zero 에 가까움. **strict 한 형식적 mediation claim 은 신중.**
5. **Pillar II / Pillar III 정의 미정.** Yu 미팅 input 대기. Pillar II = Bundang prospective? Pillar III = BCR-seq deep? 미정 상태에서 manuscript 시작 못 함.
6. **Image-DM1 / pathology-AI 어떤 활성-claim 도 금지.** Paper 2A NO-GO 후 forbidden vocabulary list. Methods 에 negative-feasibility 1-pager 만 허용.
7. **GD / Chu 2018 / DPB1*05:01 narrative 절대 금지.** Paper 4 territory.
8. **Voice-protected 6 sections 미작성.** Hook / Aim / Discussion §3.1 (반드시 PTC ≠ GD 강조) / §3.4 Limitations / Cover ¶1 / Reviewer Q9.

13. Claim boundary table

CLAIM	STATUS
Pillar 1–6 all DONE 5/4	YES
HLA-II 140% mediation (n=18)	YES – exploratory, single-cohort
TCGA generalization OR up to 5× ($p=6.4 \times 10^{-10}$)	YES
BCR clonal + TLS architecture (gene-level proxy)	YES – RNA-seq proxy only
DM1 sub-B = NBNR cluster cross-paper bridge	YES
Pillar II / Pillar III definition	TBD – Yu input 대기
Mediation replication independent cohort	PENDING (Bundang)
BCR-seq deep (CDR3 motif)	PENDING (Lim 2025 후속)
Image-DM1 / pathology-AI	FORBIDDEN (Paper 2A NO-GO)
GD / Chu 2018 / DPB1 GD narrative	FORBIDDEN (Paper 4 territory)
Voice-protected 6 sections (Hook/Aim/Disc/Lim/Cover/Q9)	UNWRITTEN – author keyboard only

I4. 데이터 / 다운로드

 **GSEA HALLMARK
2020**
top 30 enriched terms

 **GSEA KEGG 2021**
top 30 (autoimmune
dominant)

 **GSEA REACTOME
2022**
top 30

 **HLA MODULE
SCORES**
per sample (n=18)

 **8-GENE PER
GENE**
PTC vs PTC+HT
compare

 **DM1/DM2
PREDICTIONS**
GSE286332 classifier
output

 **DEG PTC+HT VS
PTC**
top 50

 **COHORT
ASSEMBLY**
Korean PTC pool
n=874 + TCGA + Chu
2018

 **PILLAR I
FOREST META**
per-allele

 **RANDOM-EFFECTS
POOLED OR**
DerSimonian-Laird

 **KOREAN SUB-
COHORT
HETEROGENEITY**
Cochran's Q + I²

 **4-SCENARIO
SENSITIVITY**
full / excl
GSE286332 / Lee /
K2

 **DM1 SUB-B VS A
DEGS**
top 50

← **PAPER 2A (H&E
NO-GO)**
Methods supplement

→ **PAPER 4 (KOREAN
GD HLA)**
Pan-Asian backlog

→ **PAPER 3 (ICI
FROZEN)**
Track A FROZEN

← **PAPER 1 (DM1
MOLECULAR)**
cross-paper bridge

← **HUB INDEX**
8-papers portfolio

Status. Paper 2B = Hashimoto-overlap PTC mechanism · Active marathon · Pillar 1–6
DONE 5/4 · Pillar II/III 정의 미정 (Yu 다음 미팅 input 대기) · Venue: JCI Insight / Cell
Rep Med.

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